Stress Assignment without Morphology in Portuguese: the qualityless vowel /0/ and the Prosody-Conditioned Extrametricaliser (PCE)[1]

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Abstract:

This paper offers an Optimality-Theoretic reinterpretation and synthesis of the author's earlier Japanese works (Makino 2010, 2012) on Portuguese stress patterns, focusing exclusively on the theoretical essentials. We put forward a prosody-only account of Portuguese word stress. The analysis reduces the Portuguese stress system to the interaction of **a qualityless vowel /0/ bearing one mora + *prosody-conditioned extrametricaliser* (PCEs) + general metrical theory formalised through OT constraints**. This /0/ is the musical "**ghost note**" — **beat** (temporal position) without **timbre** (quality) — which nevertheless shapes the **rhythm** (stress pattern): e.g. /0/ in *contador* /kont-a-dor-**0**/ vs. *contadora* /kont-a-dor-**a**/. PCEs are a class of morpheme-sized items that, **when the condition is satisfied, render the final syllable of the prosodic word extrametrical**.

Principles

- Trochaic binary footing from the right edge
- The rightmost foot functions as the head foot

Parameter: determination of the right edge

- **Value 1:** final syllable — unmarked default
- **Value 2:** penultimate syllable — marked (when a PCE satisfies the condition)

Metrification is computed strictly from phonological information, together with a single piece of non-categorial lexical information — namely, the **PCE condition**. No reference to syntactic categories or to morphological bracketing is made at any point in the stress-assignment process. This approach thus **eliminates morphology from stress computation proper** and

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unifies items such as *contador*, *contadora*, *falava*, *falávamos*, *faláveis*, *número*, *numeroso*, *numérico*, *sobre* and *após* within a single minimal mechanism.

0. Abbreviations, symbols and diacritics

PrWd = prosodic word; **Ft** = foot; **HdFt** = head foot; **PCE** = *prosody-conditioned extrametricaliser*; **σ_{final}_PCE** = the syllable whose nucleus is the final vowel of a PCE; **L** = light syllable; **H** = heavy syllable

FTBIN = Parse syllables into a binary foot; **TROCHEE** = Parse syllables into a trochaic foot, where the first syllable is strong and the remaining syllable(s) are weak; **ALIGN-R(Ft, PrWd)** = Align the right edge of each foot with the right edge of the prosodic word; **ALIGN-R(HdFt, PrWd)** = Align the right edge of the head foot with the right edge of the prosodic word; **ALIGN-R(Ft, σ_{final}_PCE)** = Align the right edge of each foot with the syllable whose nucleus is the final vowel of a PCE; **ALIGN-R(HdFt, σ_{final}_PCE)** = Align the right edge of the head foot with the syllable whose nucleus is the final vowel of a PCE; **PARSE(σ)** = Parse every syllable into a foot; ***CLASH** = Avoid a stress clash; **#WSP** = Weight-to-Stress Principle at the left edge of the prosodic word

/ / = underlying representation; **[ ]** = surface representation; **→** = derivation; **-** = morpheme boundary; **#** = word boundary; **/0/** = qualityless vowel bearing one mora; **C** = consonant; **C̩** (e.g. r̩, n̩, s̩) = syllabic consonant that arises from the association of a consonant with the mora supplied by /0/; **μ_0** = the mora supplied by /0/; **.** = syllable boundary; ˈ = primary stress; ˌ = secondary stress; **σ** = syllable; **( )** = footing

1. Introduction

While previous accounts have relied on morphological information in stress computation, the present approach **dispenses with morphology and derives stress solely from phonological information** — syllable string, syllable number, syllable weight (at the left edge) and PCE condition.

2. The qualityless vowel /0/

A **qualityless vowel /0/ that bears one mora** in underlying representations is assumed — e.g. *contador* /kont-a-dor-**0**/ (cf. *contadora* /kont-a-dor-**a**/); *jacaré* /ʒakarɛ-**0**/; *trabalhar* /trabaʎ-a-r**0**/; *trabalharmos* /trabaʎ-a-r**0**-mos/; *após* /a-pɔ**0**s/; *até* /atɛ**0**/. This vowel /0/ is the musical "**ghost note**" — **beat** (temporal position) without **timbre** (quality) — which nevertheless shapes the **rhythm** (stress pattern). It is a purely phonological entity, serving various morphological purposes: in *contador* /kont-a-dor-**0**/ and *jacaré* /ʒakarɛ-**0**/, the /0/ functions as the thematic vowel — stem vowel or *índice temático* (stem marker) — just as /a/ does in *contadora* /kont-a-dor-**a**/

(also analisable as /kont-a-dor-0-**a**/). On the other hand, it forms part of the infinitive or future subjunctive suffix /-r0/ in *trabalhar* /trabaʎ-a-r**0**/, and likewise occurs in the preposition *até* /atɛ**0**/, which does not display any internal morphological structure.

The postulation of /0/ is not an ad hoc assumption but rather a corollary deduced from a stress theory based on the metrical foot. Strong–weak alternation presupposes at least two distinct temporal positions on the time axis. Without such positions, alternation in strength cannot be realised. Conversely, distinct temporal positions do not presuppose strength. It follows that temporal positions can exist independently of strength. In other words, **distinct temporal positions are a prerequisite for strong–weak alternation**, but not vice versa.

If, metrical-theoretically, a stress pattern is defined as the *iterative repetition of the alternation between strong and weak beats along the time axis*, then both strong and weak beats presuppose distinct temporal positions. When the main stress — the most prominent strong–weak alternation — falls on *-dor* in *conta**dor***, **there must necessarily exist within *-dor* the temporal position corresponding to the weak beat paired with the strong one**. If strong and weak beats are borne by the syllable — more precisely, by the syllable nucleus — then, since the strong beat is borne by the syllabic /o/ in /-dor/ as a moraic temporal position, **the paired weak beat must be borne by the syllabic mora /0/ as a temporal position**. Thus, positing /0/ — a qualityless, syllable-nucleus-projecting mora — is required both by **the primacy of temporal positions for strong–weak alternation** and by **the very definition of the *stress pattern***.

Accordingly, if /0/ were not assumed, *-dor* would constitute a degenerate foot — which, within metrical theory, is by definition a monosyllable left over at the end of footing — and this imperfect foot would serve as the head of the prosodic word, bearing the highest prominence. This outcome would be theoretically undesirable, since footing in Portuguese is assumed to proceed from the right edge, as shown in Sections 4–6, and yet its initial foot would paradoxically be degenerate — a state of affairs for which metrical theory may offer no principled explanation. For further support, **empirical corroborations** will be presented in Section 8.

3. The prosody-conditioned extrametricaliser (PCE)

It is also assumed that there exists a class of ***prosody-conditioned extrametricalisers*** **(PCEs)**. These are **morpheme-sized items** that are **lexically specified to render the final syllable of the PrWd extrametrical** (i.e. excluded from footing), but **only when their own final vowel serves as the nucleus of the penultimate syllable of the PrWd** — e.g. *-ic-* /-ik-/ in *académ**ic**o* /akadɛm-**ik**-o/; *-vel* /-vel/ in *agradá**vel*** /a-grad-a-**vel**-0/; *numer-* /numɛr-/ in *número* /**numɛr**-o/; *-va-* /-va-/ in *trabalhá**va**mos* /trabaʎ-a-**va**-mos/; *-r-* /-r0-/ in *trabalha**r**des* /trabaʎ-a-**r0**-des/. **The membership of a morpheme in this class is independent of its morphosyntactic category**.

4. Unmarked metrification

Regardless of syntactic category or morphological structure, **iterative formation of trochaic binary feet proceeds from the final syllable of the PrWd towards the word-initial edge**, with the **rightmost foot** serving as the head and bearing **primary stress**:

possibilidade /pɔs-i-bil-i-dad-e/ → pɔ.si.bi.li.da.de → (ˌpɔ.si)(ˌbi.li)(ˈda.de) → [ˌpu.siˌβɨ.liˈðaː.ðɨ]
contador /kont-a-dor-0/ → kon.ta.do.ɽ̩ → (ˌkon.ta)(ˈdo.ɽ̩) → [ˌkõ.tɐˈðoːɾ]
contadora /kont-a-dor-a/ → kon.ta.do.ra → (ˌkon.ta)(ˈdo.ra) → [ˌkõ.tɐˈðoː.ɾɐ]
coração /korason-0/ → ko.ra.so.n̩ → (ˌko.ra)(ˈso.n̩) → [ˌku.ɾɐˈsɐ̃ũ̯]
parabéns /para-ben-0-s/ → pa.ra.ben.0s → pa.ra.ben.es → (ˌpa.ra)(ˈben.es) → [ˌpɐ.ɾɐˈβɐ̃ĩ̯ʃ]
jacaré /ʒakarɛ-0/ → ʒa.ka.rɛ.ɛ → (ˌʒa.ka)(ˈrɛ.ɛ) → [ˌʒɐ.kɐˈɾɛː]
trabalhamos /trabaʎ-a-mos/ → tra.ba.ʎa.mos → (ˌtra.ba)(ˈʎa.mos) → [ˌtɾɐ.βɐˈʎɐː.muʃ]
trabalhais /trabaʎ-a-des/ → tra.ba.ʎa.des → (ˌtra.ba)(ˈʎa.des) → [ˌtɾɐ.βɐˈʎai̯ʃ]
trabalhar /trabaʎ-a-r0/ → tra.ba.ʎa.ɽ̩ → (ˌtra.ba)(ˈʎa.ɽ̩) → [ˌtɾɐ.βɐˈʎaːɾ]
trabalhei /trabaʎ-a-i/ → tra.ba.ʎa.i → (ˌtra.ba)(ˈʎa.i) → [ˌtɾɐ.βɐˈʎɐi̯]
trabalhou /trabaʎ-a-u/ → tra.ba.ʎa.u → (ˌtra.ba)(ˈʎa.u) → [ˌtɾɐ.βɐˈʎoː]
sobre /sobre/ → so.bre → (ˈso.bre) → [ˈsoː.βɾɨ]

This suggests the following constraints in interaction: **FtBin; Trochee; Align-R(Ft, PrWd); Align-R(HdFt, PrWd)**. Note that the syllabification of /0/ is hypothetical and that various possibilities are theoretically conceivable. What is crucial from a metrical-theoretical perspective, however, is that **/0/ is not merely a so-called stray/floating mora**, but **rather an element constituting a syllable nucleus**, and therefore that the syllable containing the mora supplied by /0/ — henceforth μ_0 — is a separate syllable from those preceding and following it. **The notation /0/** is adopted precisely to indicate that it constitutes **a syllabic unit**. Note also that, whereas the syllabification of segments is assumed to refer not only to phonological but also to morphological information, the metrification of syllable strings is assumed to refer solely to phonological information.

5. Marked metrification

If the **final vowel of a PCE** serves as the **nucleus of the penultimate syllable of the PrWd** (the **PCE condition**), the same iterative formation of trochaic binary feet proceeds **from that penultimate syllable — rather than from the final one** — with the **rightmost foot** serving as the head foot and bearing **primary stress**. The final syllable is thereby excluded from metrification — that is, *extrametricalised* — and remains unstressed:

*acadé**mi**co* /akadɛm-***i*k**-o/ → a.ka.dɛ.m***i*.k**o → (ˌa.ka)(ˈdɛ.m***i***)**k**o → [ˌɐ.kɐˈðɛː.mi.ku]

*agradá**vel*** /a-grad-a-**v*e*l**-0/ → a.gra.da.**v*e*.l̩** → (ˌa.gra)(ˈda.**v*e***)**l̩** → [ˌɐ.ɣɾɐˈðaː.vɛɫ]

núm**er**o /**num*ɛ*r**-o/ → **nu**.**m*ɛ***.**r**o → (ˈ**nu**.**m*ɛ***)**r**o → [ˈnuː.mɨ.ɾu]

tún**e**l /**tun*e*l**-0/ → **tu**.**n*e*.l̩** → (ˈ**tu**.**n*e***)**l̩** → [ˈtuː.nɛɫ]

só**t**ão /**sɔ*ta*n**-o/ → **sɔ**.***ta*n**.o → (ˈ**sɔ**.***ta*n**)o → [ˈsɔː.tɐ̃ṵ̃]

*trabalhá**va**mos* /trabaʎ-a-**v*a***-mos/ → tra.ba.ʎa.**v*a***.mos → (ˌtra.ba)(ˈʎa.**v*a***)mos

→ [ˌtɾɐ.βɐˈʎaː.vɐ.muʃ]

*trabalhá**ve**is* /trabaʎ-a-**v*a***-des/ → tra.ba.ʎa.**v*a***.des → (ˌtra.ba)(ˈʎa.**v*a***)des → [ˌtɾɐ.βɐˈʎaː.vɐi̯ʃ]

*trabalha**r**des* /trabaʎ-a-**r*0***-des/ → tra.ba.ʎa.***r̩***.des → (ˌtra.ba)(ˈʎa.***r̩***)des → [ˌtɾɐ.βɐˈʎaɾ.ðɨ̯ʃ]

By contrast, if the **final vowel of a PCE** does **not function as the nucleus of the penultimate syllable of the PrWd**, **unmarked metrification**, as described in Section 4, emerges:

*ato**nic**idade* /a-tɔn-***i*k**-i-dad-e/ → a.tɔ.n***i*.s**i.da.de → (ˌa.tɔ)(ˌn***i*.s**i)(ˈda.de) → [ˌɐ.tuˌni.siˈðaː.ðɨ]

numer**oso* /**num*ɛ*r**-oz-o/ → **nu.m*ɛ*.r**o.zo → (ˌ**nu.m*ɛ)(ˈ**r**o.zo) → [ˌnu.mɨˈɾoː.zu]

*trabalha**va*** /trabaʎ-a-**v*a***/ → tra.ba.ʎa.**v*a*** → (ˌtra.ba)(ˈʎa.**v*a***) → [ˌtɾɐ.βɐˈʎaː.vɐ]

*trabalha**r*** /trabaʎ-a-**r*0***/ → tra.ba.ʎa.***r̩*** → (ˌtra.ba)(ˈʎa.***r̩***) → [ˌtɾɐ.βɐˈʎaːɾ]

When a PrWd contains **more than one PCE**, metrification proceeds from **the syllable whose nucleus is the final vowel of the PCE that satisfies the PCE condition** stated above:

numéric**o* /**num*ɛ*r-*i*k**-o/ → **nu**.**m*ɛ.**r*i***.**k**o → **nu**(ˈ**m*ɛ*.r*i***)**k**o → [nuˈmɛː.ɾi.ku]

This phenomenon, in which items such as *-ic-*, *numer-*, *-va-*, *-r* shift leftward or rightward and thereby become inactive, indicates that a PCE is a **purely prosodic/metrical unit**: it **triggers extrametricalisation only under specific prosodic conditions**. This constitutes a binary prosodic on/off mechanism, **clearly distinct from general lexical accent specification**. All of the above suggests the ranking: **ALIGN-R(Ft, σ_{final}_ PCE)** ≫ **ALIGN-R(Ft, PrWd)** and **ALIGN-R(HdFt, σ_{final}_PCE)** ≫ **ALIGN-R(HdFt, PrWd)**.

6. Stress clash avoidance (provisional)

When the metrification domain contains an **odd** number of syllables, to avoid a stress clash between a potential **word-initial unary foot** and the following **binary foot**, metrification is assumed to proceed as follows. Note that these generalisations are **provisional** and call for further investigation.

6.1 Five or more syllables: only the leftmost foot is ternary

Only the **leftmost** foot is **ternary**; **subsequent** feet remain **binary** trochees:

papelaria /papel-ari-a/ → pa.pe.la.ri.a → (ˌ**pa.pe.la**)(ˈri.a) → [ˌpɐ.pɨ.lɐˈða:.ðɨ]
computador /konput-a-dor-0/ → kon.pu.ta.do.ɽ → (ˌ**kon.pu.ta**)(ˈdo.ɽ) → [ˌkõ.pu.tɐˈðo:ɾ]
entendimento /entend-i-ment-o/ → en.ten.di.men.to → (ˌ**en.ten.di**)(ˈmen.to) → [ˌẽ.tẽ.diˈẽ:.tu]

The constraint observations and violations of the winner and the losers are as follows:

Winner #(ˌσσσ)…#: NO *CLASH violation; **<u>NO PARSE(σ) violation</u>**; **ONE FTBIN violation**
Loser #σ(ˌσσ)…#: NO *CLASH violation; **ONE PARSE(σ) violation**; **<u>NO FTBIN violation</u>**
∴ **PARSE(σ) ≫ FTBIN**

Winner #(ˌσσσ)…#: **<u>NO *CLASH violation</u>**; NO PARSE(σ) violation; ONE FTBIN violation
Loser #(ˌσ)(ˌσσ)…#: **<u>ONE *CLASH violation</u>**; NO PARSE(σ) violation; ONE FTBIN violation
∴ ***CLASH** decides (tie-breaking).

This suggests the following constraints: **PARSE(σ) ≫ FTBIN**; ***CLASH**

6.2 Initial L-H exception: initial light syllable unfooted

If a **light initial syllable is immediately followed by a heavy syllable**, it remains **unfooted**:

levantamento /levant-a-ment-o/ → **le**.van.ta.men.to → **le**(ˌvan.ta)(ˈmen.to) → [lɨˌvɐ̃.taˈmẽ:.tu]

Note that optional unary footing of the initial light syllable is not excluded, as in *levantamento* /levant-a-ment-o/ → **le**.van.ta.men.to → (ˌ**le**)(ˌvan.ta)(ˈmen.to) → [ˌlɨˌvɐ̃.taˈmẽ:.tu].
The constraint observations and violations are as follows:

Winner # L(ˌH.σ)…#:
NO *CLASH violation; **ONE PARSE(σ) violation**; NO FTBIN violation; **<u>NO #WSP violation</u>**
Loser # (ˌL.H.σ)…#:
NO *CLASH violation; **<u>NO PARSE(σ) violation</u>**; ONE FTBIN violation; **ONE #WSP violation**
∴ **#WSP ≫ PARSE(σ)** — since PARSE(σ) ≫ FTBIN, as inferred in Section 6.1, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

Winner #L(ˌH.σ)…#:

<u>NO *CLASH violation</u>; **ONE PARSE(σ) violation**; NO FTBIN violation; NO #WSP violation

Loser #(ˌL)(ˌH.σ)…#:

ONE *CLASH violation; **<u>NO PARSE(σ) violation</u>**; ONE FTBIN violation; NO #WSP violation

∴ ***CLASH ≫ PARSE(σ)** — since PARSE(σ) ≫ FTBIN, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

This suggests the following constraint ranking: ***CLASH, #WSP ≫ PARSE(σ) ≫ FTBIN**

6.3 Three syllables: initial syllable unfooted

The **initial** syllable is **not footed**:

pessoa /peso-a/ → **pe**.so.a → **pe**(ˈso.a) → [**pɨ**ˈsoː.ɐ]
doutores /dout-or-0-s/ → **doṷ**.to.r0s → **doṷ**.to.res → **doṷ**(ˈto.res) → [**do**ˈtoː.ɾɨʃ]
pequeno /peken-o/ → **pe**.ke.no → **pe**(ˈke.no) → [**pɨ**ˈkeː.nu]
levar /lɛv-a-r0/ → **lɛ**.va.ɽ → **lɛ**(ˈva.ɽ) → [**lɨ**ˈvaːɾ]
até /atɛ0/ → **a**.tɛ.ɛ → **a**(ˈtɛ.ɛ) → [**ɐ**ˈtɛː]
após /apɔ0s/ → **a**.pɔ.ɔs → **a**(ˈpɔ.ɔs) → [**ɐ**ˈpɔːʃ]

Note that optional unary footing of the initial syllable, as in *levar* /lɛv-a-r0/ → lɛ.va.ɽ → (lɛ)(ˈva.ɽ) → [ˌlɨˈvaːɾ], is not excluded. The constraint observations and violations of the winner candidate and the loser candidates are as follows:

Winner #σ(ˈσσ)#: **<u>NO *CLASH violation</u>**; **ONE PARSE(σ) violation**; NO FTBIN violation
Loser #(ˌσ)(ˈσσ)#: **ONE *CLASH violation**; **<u>NO PARSE(σ) violation</u>**; ONE FTBIN violation
∴ ***CLASH ≫ PARSE(σ)** — since PARSE(σ) ≫ FTBIN, the evaluation by FTBIN does not affect the evaluation by PARSE(σ).

This also suggests the following constraint ranking: ***CLASH, #WSP ≫ PARSE(σ) ≫ FTBIN**

7. The constraint set (provisional)

From the discussion in Sections 4–6, the following set of constraints and their rankings are inferred:

***CLASH, #WSP ≫ PARSE(σ) ≫ FTBIN;**

TROCHEE;

ALIGN-R(Ft, σ_{final}_PCE) ≫ ALIGN-R(Ft, PrWd);

ALIGN-R(HdFt, σ_{final}_PCE) ≫ ALIGN-R(HdFt, PrWd)

8. Plausibility of the postulation of /0/

8.1 Historical corroboration: from Latin to Portuguese

Comparing the Portuguese nouns *amador* /am-a-dor-**0**/ and *razão* /razon-**0**/ with their Latin etymons *amatorem* /amā-t-ōr-**e**-m/ and *ratiōnem* /rat-iōn-**e**-m/ shows that the Portuguese /0/ corresponds to the Latin segment /e/. Many other examples of this kind can also be found. This suggests that, in this case, the Portuguese **/0/ was established as the result of the Latin /e/ having its quality deleted while its quantity was preserved**, thereby **preventing main-stress displacement through the preservation of the head trochaic foot**.

8.2 Synchronic corroboration: Plural in *-es* [-ɨʃ] as the /-0s/ realisation

We analyse plural nouns and adjectives ending in *-**es***, such as *contador**es***, as /kont-a-dor-**0**-s/, with the thematic vowel /-0/. In the singular *contador* /kont-a-dor-**0**/, μ_0 is taken to be associated with the final rhotic /-r/ during syllabification, yielding (ˌkon.ta)(ˈdo.ɾ̩) when metrified, which surfaces as [ˌkõ.tɐˈðoːɾ] through resyllabification. In the plural, on the other hand, the /0/ in the sequence /-C**0**s/# ultimately surfaces as [ɨ] through the linking of μ_0 to the epenthesised vocalic quality, yielding the surface form [ˌkõ.tɐˈðoː.ɾ**ɨ**ʃ] from the underlying representation /kont-a-dor-**0**-s/. This analysis avoids positing an underlying /e/ in the plural — as in /kont-a-dor-**e**-s/ — which would have to be deleted in the singular /kont-a-dor-**e**/ at the surface level. Such a move would be incompatible with the many lexical items that retain final /-e/ [-ɨ] — e.g. *árvor**e***, *ar**e***, *hectar**e***, *Zêzer**e***. Note finally that, by **positing /-0/ alongside /-a/, /-e/, /-o/ as a thematic vowel**, **all inflexional words** — namely, nouns, adjectives and verbs — can be analysed as **sharing the common structure [[[radical] thematic vowel] inflexional suffixes]**, while at the same time allowing **the stress pattern to be derived solely on the basis of phonological information.**

8.3 Northern EP corroboration: paragoge

In Northern varieties, words ending in /-r**0**/ or /-l**0**/ may surface as [-ɾ**i**] ~ [-ɾ**ɨ**] or [-l**i**] ~ [-l**ɨ**]. For instance, the infinitive *comer* /kɔm-e-r**0**/ may surface as [kuˈmeː.ɾ**i**] or [kuˈmeː.ɾ**ɨ**], with paragogic [i] or [ɨ], rather than as the standard [kuˈmeːɾ]. At the surface level, μ_0 is associated with [ɾ] in the standard variety, whereas in those Northern varieties **μ_0 may instead be associated with the epenthetic quality [i] or [ɨ]**, with **the stress pattern remaining unchanged**. In other words, the "ghost note" /0/ thus occasionally becomes associated with *inserted* vocalic timbre, thereby becoming a "full note" [i] or [ɨ], yet without altering the rhythm. This indicates that **μ_0, irrespective of its association with any quality — consonantal or vocalic — constitutes the weak beat in a strong–weak alternation** and that this unit is not a mere stray/floating mora, but an element that projects a syllable nucleus.

8.4 Cross-Romance corroboration: Portuguese vs. Spanish

The main stress position of the future subjunctive forms *trabalharmos* and *trabalhar* can be determined solely on the basis of phonological information, by analysing the **future subjunctive suffix *-r*** as the **PCE /-r0/ with /0/**:

trabalharmos /trabaʎ-a-**r0**-mos/ → tra.ba.ʎa.**r̩**.mos → (ˌtra.ba)(ˈʎa.**r̩**)mos
trabalhar /trabaʎ-a-**r0/** → tra.ba.ʎa.**r̩** → (ˌtra.ba)(ˈʎa.**r̩**)

This future subjunctive suffix is ***-re* /-re/ with /e/** in Spanish, which, like -r /-r0/ in Portuguese, functions as a PCE and renders the final syllable of the PrWd extrametrical:

trabajáremos /trabax-a-**re**-mos/ → tra.ba.xa.**re**.mos → (ˌtra.ba.)(ˈxa.**re**)mos
trabajare /trabax-a-**re/** → tra.ba.xa.**re** → (ˌtra.ba.)(ˈxa.**re**)

From this, it becomes evident that, **in stress computation, the Portuguese "ghost note" /0/ functions in exactly the same way as the Spanish "full note" /e/**. Furthermore, the stress patterns of the Portuguese noun *coração* and verb *falar* and their Spanish cognates *corazón* and *hablar* can likewise be **derived in parallel by positing /0/**:

coração /korason-**0/** → ko.ra.so.n̩ → (ˌko.ra)(ˈso.n̩) → [ˌku.ɾɐˈsɐ̃ũ̯]
corazón /koraθon-**0/** → ko.ra.θo.n̩ → (ˌko.ra)(ˈθo.n̩) → [ˌko.ɾaˈθɔn]
falar /fal-a-r**0/** → fa.la.r̩ → fa(ˈla.r̩) → [fɐˈlaːɾ]
hablar /habl-a-r**0/** → a.bla.r̩ → a(ˈbla.r̩) → [aˈβlaɾ]

The difference between these two languages lies primarily in the degree of complexity of the phonological processes that apply after stress assignment, from the intermediate representations (ˌko.ra)(ˈso.n̩) and (ˌko.ra)(ˈθo.n̩), fa(ˈla.r̩) and a(ˈbla.r̩), respectively. Thus, **the model proposed**

in this paper may be considered extendable to Spanish as well.

8.5 Theoretical parsimony: economy of representation

Under the theoretical framework of this paper, the stress-pattern opposition between the quasi-minimal pair *tam**bém*** and ***lam**bem* can be easily explained by the final /0/ in *também* /tan-ben-**0**/, which is absent from *lambem* /lanb-e-n/:

também /tan-ben**0**/ → tan.be.n̩ → tan(ˈbe.n̩) → [tɐ̃ˈbɐ̃ȷ̃]
lambem /lanb-e-n/ → lan.ben → (ˈlan.ben) → [ˈlɐ̃ː.bɐ̃ȷ̃]

Accordingly, **the stress-pattern difference can be attributed solely to their segmental sequences, in terms of temporal profile**. Without positing /0/, by contrast, one would additionally have to assign a lexically specified accent to *também*, thereby requiring two types of information in the lexicon: segmental and lexical-accentual. Moreover, **the mere application of the Weight-to-Stress Principle to the surface forms also fails**, since in both *também* [tɐ̃ˈ**bɐ̃ȷ̃**] and *lambem* [ˈlɐ̃ː.**bɐ̃ȷ̃**] the final syllable [-bɐ̃ȷ̃] is a heavy syllable with two moras, so WSP cannot distinguish them. Therefore, **lexical specification would be required** once again in this case.

9. Orthographic corroboration

When **marked metrification** emerges due to a PCE, the stressed nucleus is **typically indicated orthographically** by an acute accent or a circumflex accent, since the outcome **deviates from the default stress position** — e.g. *âmbar*, *côdea*, *número*, *pêssego*, *sótão*, *túnel*, *académico*, *agradável*, *trabalhávamos*, *trabalháveis*. This close correlation suggests that **speakers' prosodic intuitions** about **stress location** are reflected in the standard orthography, thereby providing independent support for the analysis.

10. Conclusion

Thus, Portuguese stress can be fully accounted for by **the interaction of:**

- **The qualityless vowel /0/ bearing one mora**
- **Metrically marked items (PCEs)**
- **General metrical theory formalised through OT constraints**

without recourse to morphologically-driven stress assignment. The principles and parameter of primary stress assignment in Portuguese are as follows:

Principles

- Trochaic binary footing from the right edge
- The rightmost foot functions as the head foot

Parameter: determination of the right edge

- **Value 1:** final syllable — unmarked default
- **Value 2:** penultimate syllable — marked (when a PCE satisfies the condition)

In this way, Portuguese stress patterns are reduced to two possibilities in terms of primary stress placement, yielding a relatively simple system. However, these patterns are superimposed on a wide variety of morphological structures, and it is the attempt to describe stress assignment from the perspective of those morphological structures that appears to have contributed to the observed complexity.

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